St Ignatius' College
Riverview

2023

HIGHER
SCHOOL
CERTIFICATE
ASSESSMENT TASK 1
Topic Test

Business Studies

Task 1

Task Results

Section	Marks	Cohort Average
Multiple choice	10	7.7
Question 11	10	6.9
Question 12	10	6.7
Question 13	10	6.4
Hand in Task	5	4.8
Total	45	32.5

Section I

Question	Answer
1	B
2	A
3	A
4	D
5	D
6	A
7	C
8	C
9	A
10	C

Section II

Question 11 (10 Marks)

Snowy Ice-Creams Pty Ltd is opening a new store in the city. They plan on purchasing machinery to make all products on site. They will be selling 30 different flavours of ice-cream varieties as well as several cake desserts.

(a) Outline the impact that variety will have on the operations processes for this business.

Criteria	*Marks*
Sketch in general terms the impact of variety on the operations process of Snowy Ice-creams Pty Ltd	2
May reference an impact of variety on the operations process of Snowy Ice-creams Pty Ltd	1

Answer could include

- Definition of key term

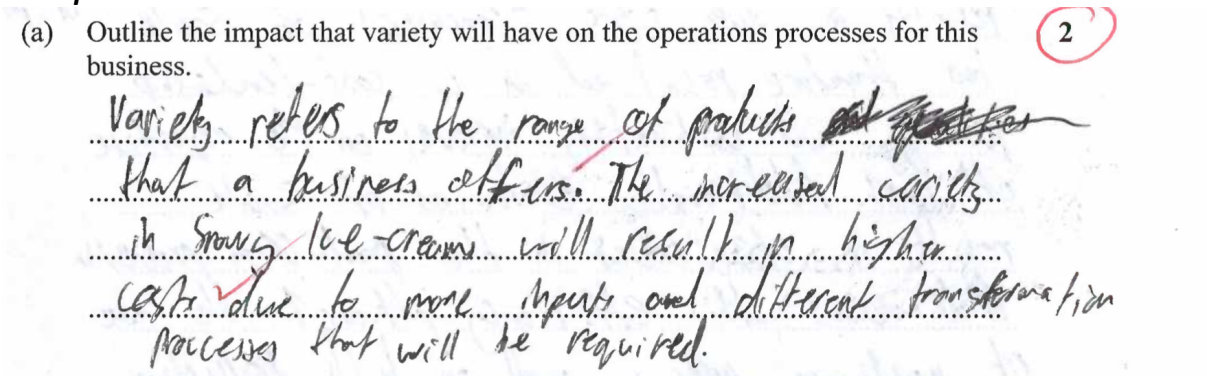
Plus, any one of the following:

- Cost increase
- Storage
- Use of technology
- Increased training
- Increased resources
- Greater choice to customers
- Impact probability – positive and negative
- Widen target market
- Increased customer base
- Greater sales

Variety refers to the range of products made by the business. With a wide variety of product being served the business is able to offer greater choice to customers. The variation in the transformation process may include new machinery, systems and procedures. This will cost the business money in terms of training and resources, therefore impacting profitability. However, this will enable the business to have a wider target market, leading to more customers, greater sales and improved profitability.

Exemplar

(a) Outline the impact that variety will have on the operations processes for this business. **2**


Variety refers to the range of products that a business offers. The increased variety in Snowy Ice-creams will result in higher costs due to more inputs and different transformation processes that will be required.

(b) Describe how the operations of this business is interdependent with human resources.

Criteria	*Marks*
Relates cause and effect the interdependence between operations and HR.	4
Provides characteristics and features of the interdependence between operations and HR.	3
Sketches in general terms the interdependence between operations and HR.	2
Recognises and names interdependence between operations and HR.	1

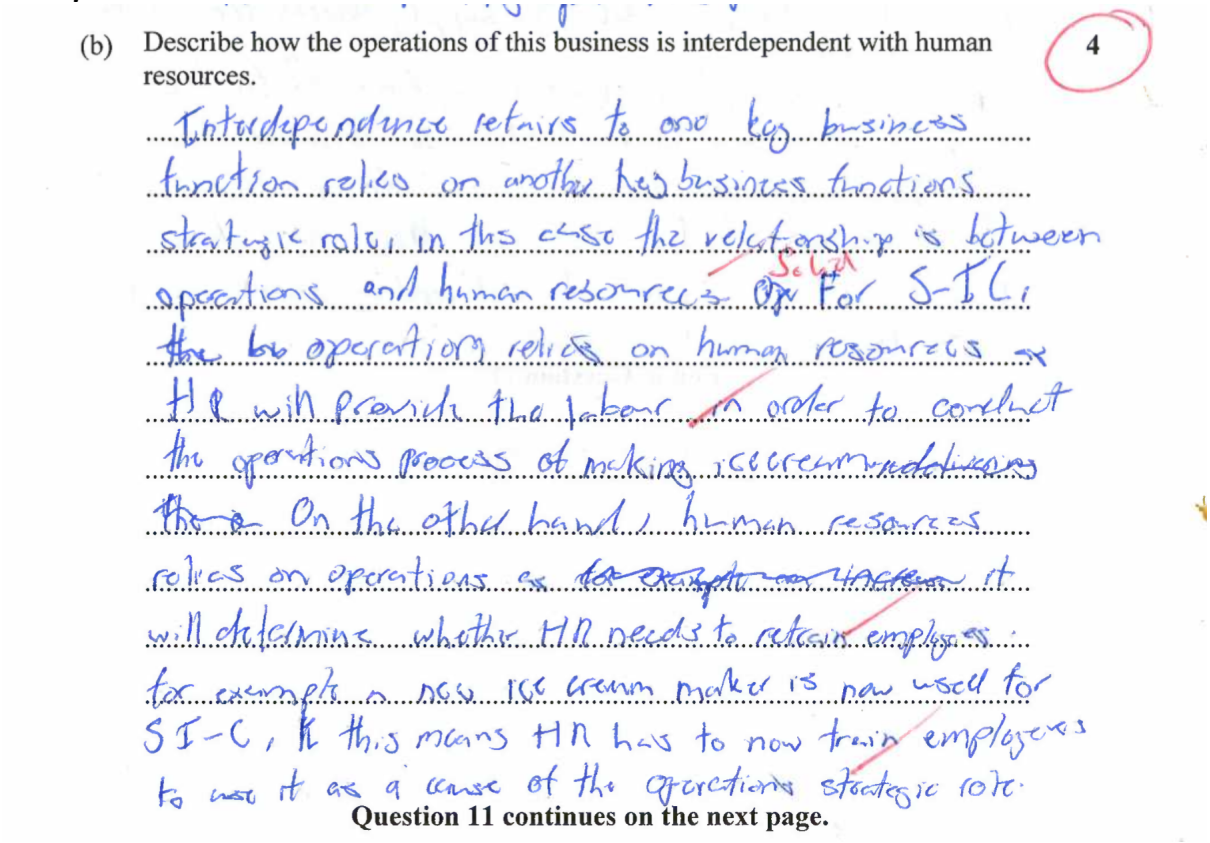
Answer could include

OPS - Inform skills and experience needed for the role.
- Job specification
- Number of employees required
- Training requirements

HR - Provide correct number of employees
- source employees with correct skill set
- Provide training
- Any elements of the recruitment process

Interdependence refers to the relationship between the core business functions, including the relationship between operations and HR. Needed to relate cause and effect of interdependence of operations on HR AND HR on operations. Operations managers will inform human resources of the skills and experience it needs its workers to have. Human resources will ensure that enough employees with the appropriate skills are available for the operations functions and provide training if needed. The human resources manager will use their leadership style and rewards to ensure quality work is done by the employees in the operations functions.

Exemplar



(b) Describe how the operations of this business is interdependent with human resources. 4

Interdependence refers to one key business function relies on another key business functions strategic roles in this case the relationship is between operations and human resources. Select For S-I C, the key operations relies on human resources as HR will provide the labour in order to conduct the operations process of making ice cream ~~and delivering them~~. On the other hand, human resources relies on operations as ~~for example when HR~~ it will determine whether HR needs to retain employees for example a new ice cream maker is now used for S I-C, if this means HR has to now train employees to use it as a cause of the operations strategic role.

Question 11 continues on the next page.

(c) With reference to a business you have studied, explain how the use of an operations strategy has improved the performance objectives of cost and speed.

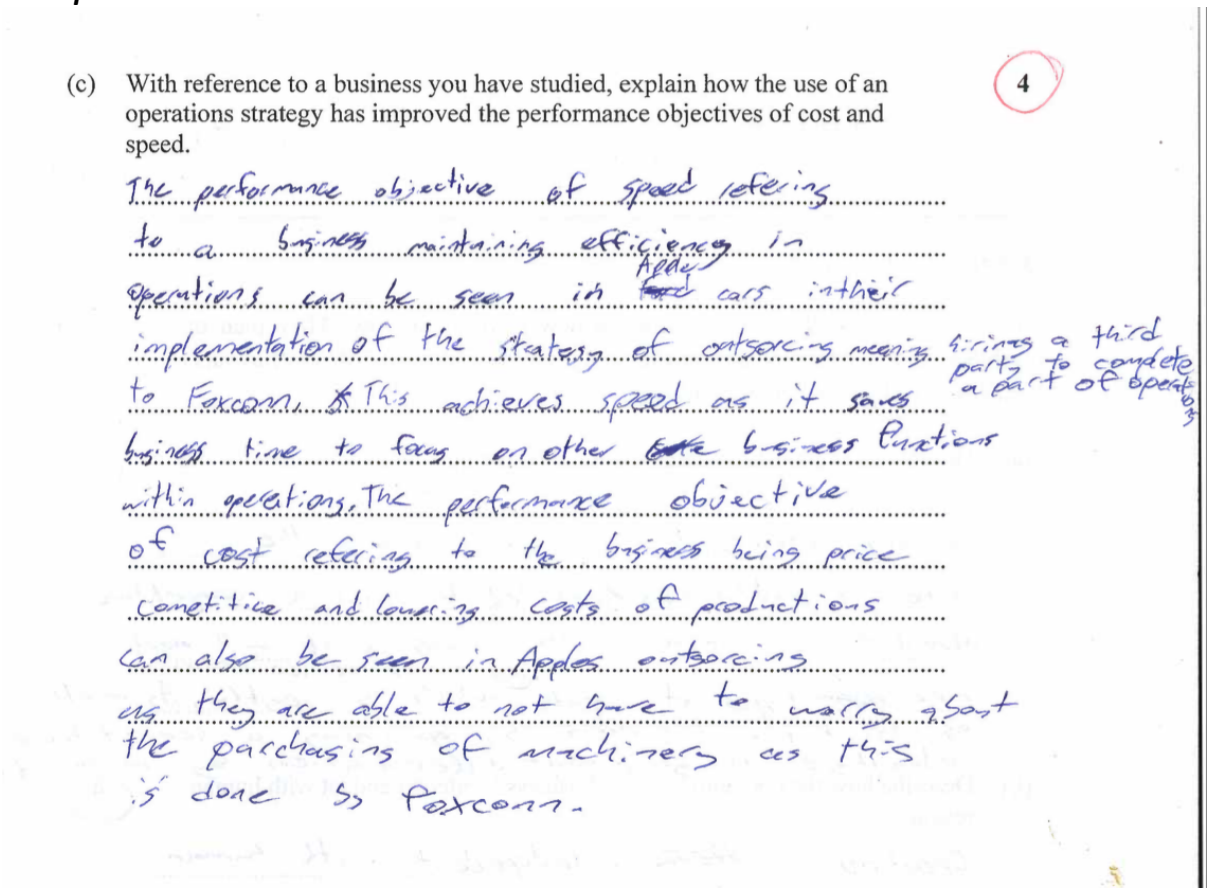
Criteria	*Mark*
Provides cause and effect in a case study addressing operations strategy to improve performance	4
Provides characteristics and features in a case study addressing operations strategy to improve performance	3
Provides theoretical cause and effect addressing operations strategy to improve performance	2
Provides characteristics and features addressing operations strategy to improve performance	1

Answer could include

- *New product or service design and development*
- *SCM - Logistics, e-commerce, Global sourcing etc*
- *Outsourcing*
- *Technology*
- *Inventory management*
- *Quality Management*

Outsourcing is the contracting out of a non-core business activity. A business may wish to focus on its main activity, and so organises another business to provide a support service such as transport, supply chain management, and producing components or supplying materials. By outsourcing, a business can free up resources to invest in the core business activities. Apple uses outsourcing in an effective way, by contracting the production of Apple products to Foxxcon, Apple can focus on both R &D and marketing, which Apple excel at. Apple can also achieve economies of scale through outsourcing, as large qualities of product will be manufactured, driving down the raw material cost through bulk buying and Foxconn running large production volumes.

Exemplar



(c) With reference to a business you have studied, explain how the use of an operations strategy has improved the performance objectives of cost and speed. 4

The performance objective of speed refering to a business maintaining efficiency in operations can be seen in Apple cars in their implementation of the strategy of outsourcing meaning hiring a third party to complete a part of operations to Foxconn. This achieves speed as it saves business time to focus on other business functions within operations. The performance objective of cost refering to the business being price competitive and lowering costs of productions can also be seen in Apples outsourcing as they are able to not have to worry about the purchasing of machinery as this is done by Foxconn.

Question 12 (10 Marks)

Optics is a manufacturer of frames for glasses including everyday reading frames and sports frames. They have recently introduced new technology to reduce the amount of plastic in each pair of their new product line by 25%.

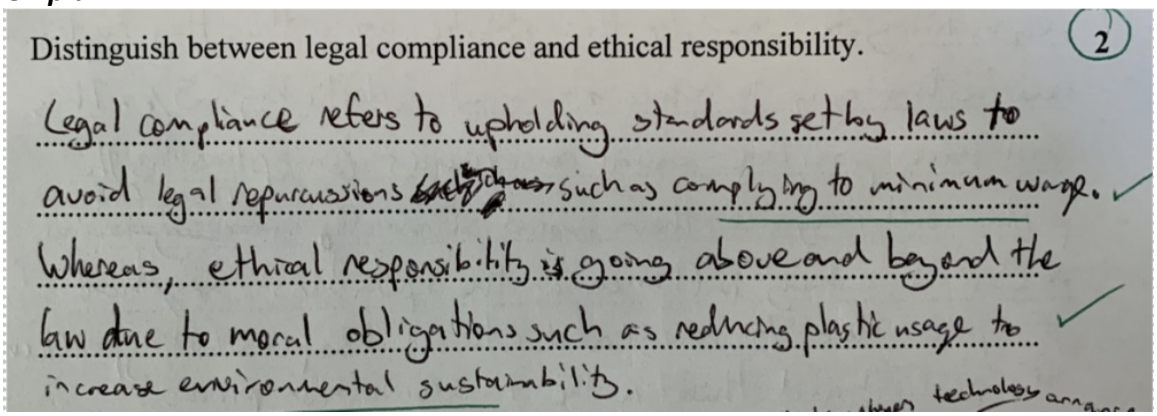
(a) Distinguish between legal compliance and ethical responsibility.

Criteria	*Marks*
Demonstrates a clear understanding of the difference between legal compliance and ethical responsibly	2
Provides some relevant information about legal compliance OR ethical responsibility	1

Answer could include

Legal compliance is when a business must adhere to the laws set out by the government. Failure to comply could result in fines or imprisonment. For example, the Fair Work Act 2009 which sets out working rights. Whereas, ethical responsibility is when a business goes above and beyond the law to engage in social or environmental causes. For example, optics giving out free reading glasses to kids.

Exemplar


Distinguish between legal compliance and ethical responsibility. 2

Legal compliance refers to upholding standards set by laws to avoid legal repurcussions such as complying to minimum wage.

Whereas, ethical responsibility is going above and beyond the law due to moral obligations such as reducing plastic usage to increase environmental sustainbility.

(b) How does the new product line impact task design and process layout in their operations?

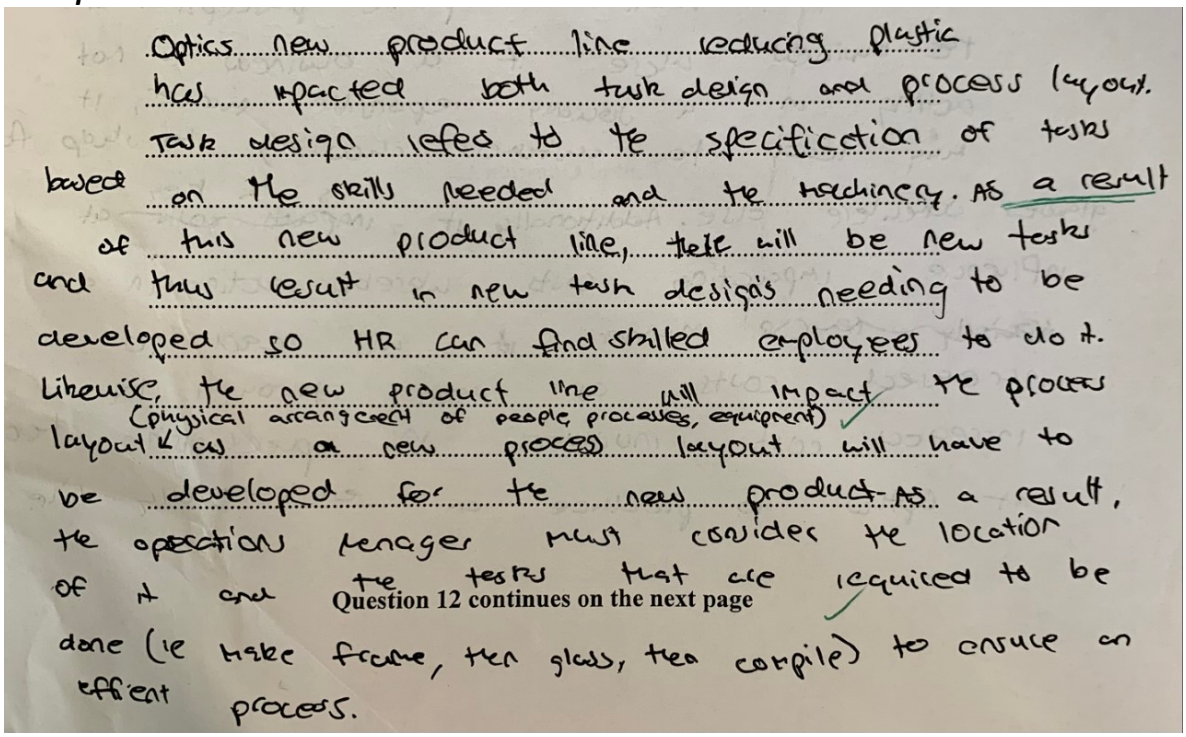
Criteria	*Marks*
Provides detailed ways in which the new product line impacts task design **AND** process layout	4
Provides ways in which the new product line impacts task design and process layout	3
Provides ways in which the new product line impacts task design **OR** process layout Provides characteristics and features of task design and process layout	2

Provides some relevant information about task design and process layout	1

Answer could include

The new product line includes new technology to help reduce plastic by 25%. Optics will need firstly to review their task design; this is the planning of tasks that need to be performed and what skills are required. For Optics, they may need to reconsider where the technology is located in the factory, and any additional training staff may need to operate the machine. In addition, process layout, which is how equipment, processes and people are laid out in production will change. For example, Optics may arrange the physical layout with the new technology at the start to produce the frames and then use a product layout to split reading and sports frames for specialist employees.

Exemplar



Optics new product line reducing plastic has impacted both task design and process layout. Task design refers to the specification of tasks based on the skills needed and the machinery. As a result of this new product line, there will be new tasks and this result in new task designs needing to be developed so HR can find skilled employees to do it. Likewise, the new product line will impact the process layout (physical arrangement of people, processes, equipment) ✓ as a new process layout will have to be developed for the new product. As a result, the operation manager must consider the location of it and the tasks that are required ✓ to be done (ie make frame, then glass, then compile) to ensure an effient process.

Question 12 continues on the next page

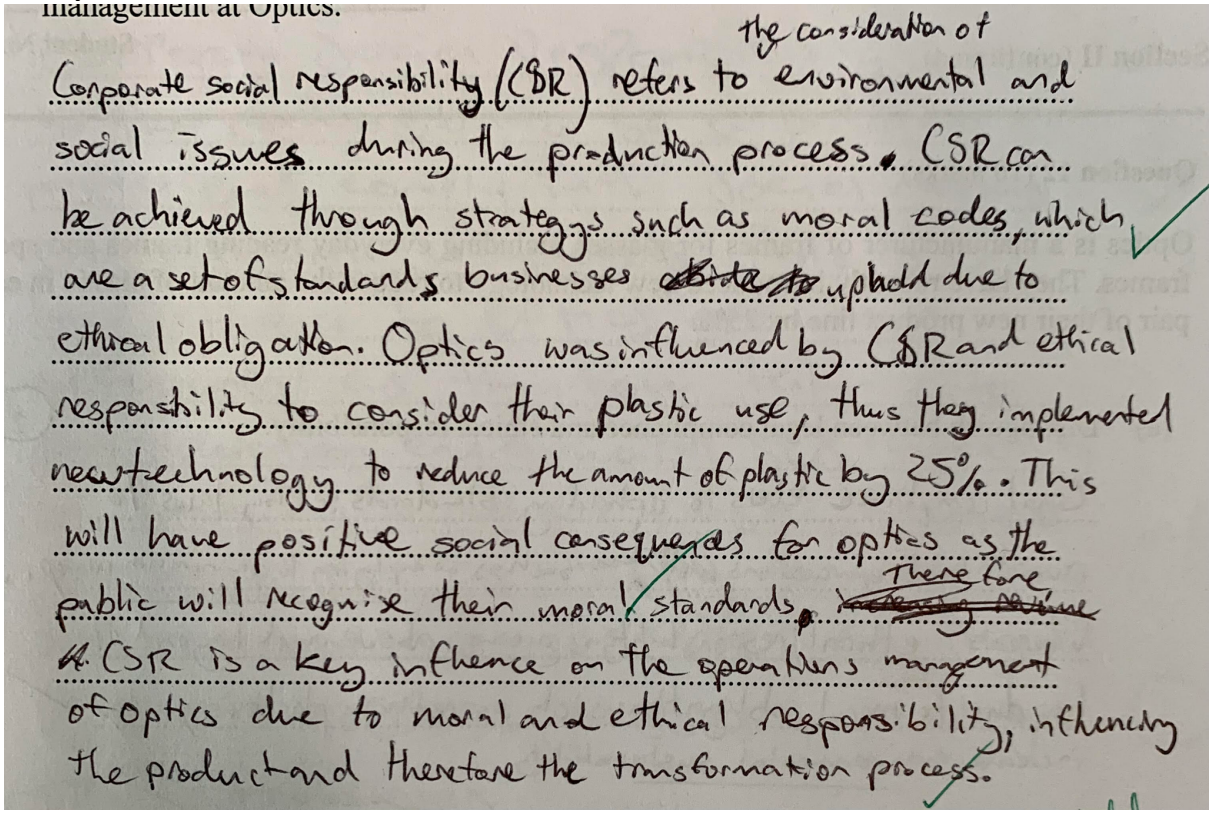
(c) Explain why corporate social responsibility is a key influence on operations management at Optics.

Criteria	*Mark*
Provides a detailed explanation of why CSR is a key influence on operations management at Optics	4
Provides an explanation of why CSR is an influence on operations management at Optics	3
Provides some relevant information about CSR linked to Optics	2
Provides some relevant information about CSR	1

Answer could include

Corporate Social Responsibility (CSR) is when a business goes above and beyond the law to engage in social activities. At Optics CSR is a key influence on operations management as they have recently installed new technology to reduce plastic usage by 25%. This impacts their reputation as a sustainable business and therefore needs careful consideration by the operations management. Optics are significantly influenced by CSR and will need to review its material inputs and the transformation process to ensure they are engaging with sustainable practices throughout the business to keep up with evolving consumer expectations.

Exemplar


management at Optics.

Corporate social responsibility (CSR) refers to the consideration of environmental and social issues during the production process. CSR can be achieved through strategys such as moral codes, which are a set of standards businesses ~~abide to~~ uphold due to ethical obligation. Optics was influenced by CSR and ethical responsibility to consider their plastic use, thus they implemented new technology to reduce the amount of plastic by 25%. This will have positive social consequences for optics as the public will recognise their moral standards. ~~Therefore~~ Therefore CSR is a key influence on the operations management of optics due to moral and ethical responsibility, influencing the product and therefore the transformation process.

Question 13 (10 Marks)

Mike's Mattress Co. manufactures high quality custom mattresses. They have provided the following data about their operations process:

	Planned Results	*Actual Results*
Number of defects (per 100 units)	5	10
Number of mattress (output per day)	200	230
Sales (per month)	$20,000	$15,000
Storage costs (per month)	$1,500	$4,000

(a) Recommend an appropriate inventory management strategy for the Mattresses at Mike's Mattress Co.

Criteria	*Marks*
Provide reasons in favour of an appropriate inventory management system for Mike's Mattress Co.	2
Sketches in general terms an appropriate inventory management system for Mike's Mattress Co.	1

Answer could include
LIFO: refers to the arrangement where goods which are last into the business are positioned so they are the first goods purchased by customers – typically for non perishable goods
JIT: A process of acquiring inputs to production just in time for entry into production processes which therefore reduces capital invested in inventory of inputs and storage cost.
FIFO was not really suitable for this business.

- this question was a recommend question which required a reason for your choice not just a definition of an inventory management system
    - e.g. Just in time method would reduce the stock level that would need to be stored thus reducing the storage costs that are higher than average.

Exemplar

(a) Recommend an appropriate inventory management strategy for the Mattresses at Mike's Mattress Co. 2

Just in time is a management strategy where ~~production~~ finished goods are made just in time for consumer demand, this would be an appropriate strategy as there is limited storage required which would decrease the businesses high storage costs, 2 ✓

(b) Using TWO examples from the table above, explain the importance of monitoring and controlling for operations management for this business.

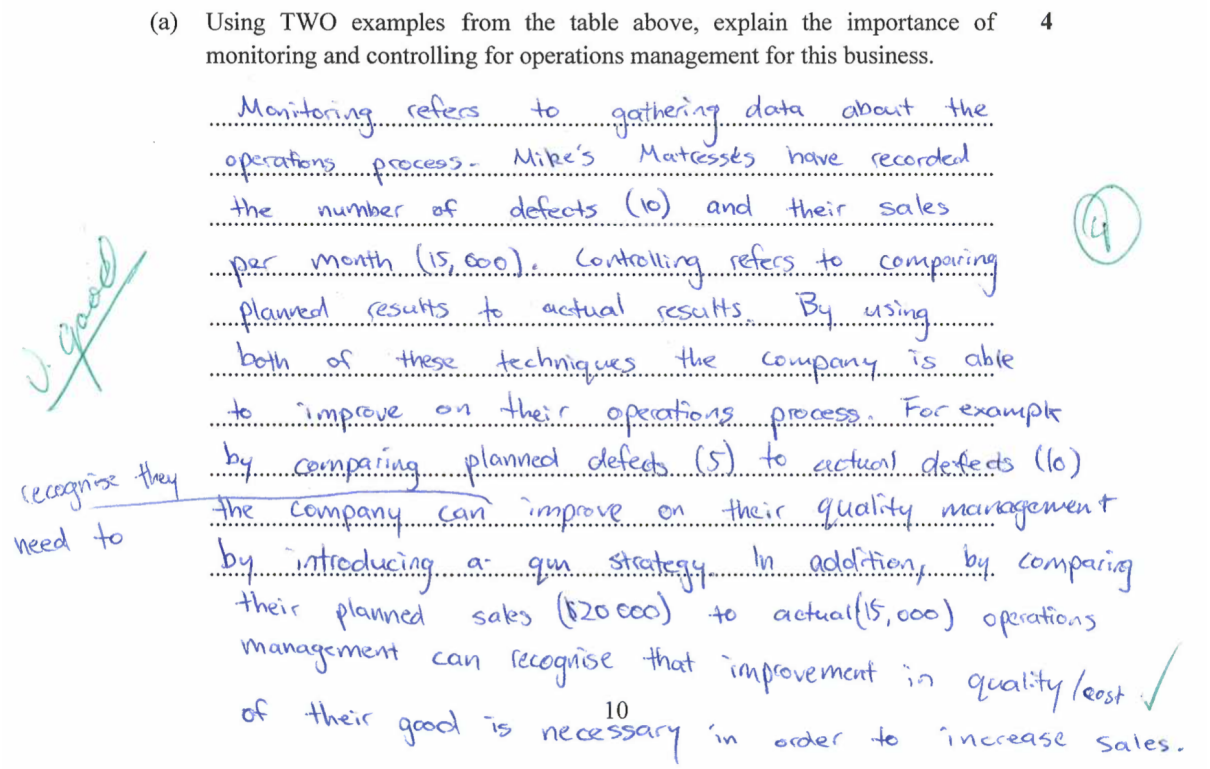
Criteria	*Marks*
Provide clear cause and effect of the importance of monitoring and controlling with the use of TWO examples from the table for this business	4
Provide somecause and effect of the importance of monitoring and controlling with the use of example(s) from the table for this business	3
Sketches in general terms the importance of monitoring and controlling with the use of example(s) from the table for this business	2
May identify some components of monitoring and controlling for this business.	1

Answer could include
Definitions: *Monitoring*: the collection of data ***Controlling***: the comparing of planned and actual data to determine if changes need to be made.

Monitoring (the collection of key operational data) and controlling (the comparison of planned and actual data) is essential in the operational management of this business. The first aspect that would be the number of defects per 100 units made. The collected data through the monitoring process recorded 10 defects in comparison to the planned data of 5 defects per 100 units, indicating that there is double to amount of defects. Through the use of monitoring and controlling Mike's Matressed could determine that corrective action such as a higher level of quality management or a reduction in overall output (as output has also increased). Additionally, the use of monitoring and controlling is important to this business when considering storage costs. The recorded data (through monitoring) indicates a cost of $4000 when compared to the planned storage costs of $1500. Through considering monitoring and controlling corrective action such as utilisation of JIT inventory management system or use of alternate storage methods could be utilised.

Exemplar on the next page

Exemplar



(a) Using TWO examples from the table above, explain the importance of monitoring and controlling for operations management for this business. 4

Monitoring refers to gathering data about the operations process. Mike's Matresses have recorded the number of defects (10) and their sales per month (15,000). Controlling refers to comparing planned results to actual results. By using both of these techniques the company is able to improve on their operations process. For example by comparing planned defects (5) to actual defects (10) the company can improve on their quality management by introducing a qm strategy. In addition, by comparing their planned sales ($20 000) to actual (15,000) operations management can recognise that improvement in quality/cost of their good is necessary in order to increase sales.

10

good

recognise they need to

(4)

This business is considering increasing their warranty from 5 years to 10 years.

(c) Explain the impact of the proposed change in the warranty for operations.

Criteria	*Mark*
Provides detailed cause and effect of the impact(s) of the proposed change of warranty for operations	4
Provides some cause and effect of the impact(s) of the proposed change of warranty for operations of Mike's Matresses	3
Provides characteristics and features of the impact of the proposed change of the warranty for operation of Mike's Matresses	2
May identify some components of an impact of the proposed change of the warranty for operations of Mike's Matresses	1

Answer could include

A warranty is a contractual commitment by the provider of a good or service to correct defects in the good or service for a specified period following purchase. The change of warranty from 5 to 10 years would have significant implications on the operations of Mike's Mattresses as they would need to provide a product that meets the warranty period. The quality of the product would need to improve, with potential improvements in inputs (e.g. springs, foam and material), aspects of transformation process such as use of leading edge machinery to ensure high quality outputs. This would result in a higher quality output that would meet the new warranty period. In addition, the relationship between quality and cost could be seen in the operations for Mike's Mattress. The utilisation of high-quality inputs and investment into new technologies would increase production costs for the business. The change in production costs could reduce profit margins for the business or have implication for marketing in determining the price for consumers.

- Other answers were considered.

Exemplar

This business is considering increasing their warranty from 5 years to 10 years.

(c) Explain the impact of the proposed change in the warranty for operations. **4**

Warranty is a quality guarantee made by a business providing the customer with assurance in the high quality of its product. Increased warranty period should result in increased quality control of the business which can be measured by the number of warranty claims, customer feedback and defects/100 units produced. Increased quality of the mattresses will be more expensive for the transformation process of the business as the inputs (materials) will have to be of a higher and more expensive quality. The operations of the transformation process may also have to be increased, potentially developing new technologies (leading edge) /implementing established technology to help achieve this higher quality output that is desired. The operations process may also be more time consuming and require increased transforming resources (human resources and facilities) to help achieve the quality that is associated with large warranty periods so that minimal warranty claims are made, which will increase brand image and customer loyalty creating increase repeat customers and higher revenue.

End of Question 13